

Gary Goh Shing Wee

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SUMMARY

Data Scientist with 8+ years in GenAI, machine learning, and data platform engineering, covering the full data function end to end: data engineering, analytics, data science, and ML Ops deployment. I architect governed data warehouses, ship LLM and agentic systems into production, and support data-driven decision making across management and fundraising. I embed AI and data governance into everything I build, driving measurable revenue, cost savings, and organisation-wide AI adoption.

SKILLS

GenAI / ML: LangChain, LangGraph, MCP, RAG, agents, Claude Code, TensorFlow, PyTorch, Strands, scikit-learn, MLflow

Data & Platform: Databricks, Apache Spark/Hive/Hadoop, AWS DMS (CDC), Unity Catalog, ETL pipeline design

Governance: AI governance, data governance, access control (Unity Catalog), PII ringfencing, read-only guardrails

Languages: Python, SQL, Bash, JavaScript, C/C++/C#, R, VBA, HTML, CSS, LaTeX

Cloud: AWS (S3, EC2, Lambda, SageMaker, Bedrock, AgentCore), GCP (GCS, Vertex AI)

Visualization: Grafana, Superset, Metabase, Lakeview, Power BI, Tableau, Matplotlib, D3.js

Databases: MySQL, Postgres, MSSQL, MongoDB, Databricks

EXPERIENCE

Data Scientist | WhiteCoat (SEA digital health platform), Singapore May 2024 – Present

- **Business Impact:** Drove regional data science initiatives (SG/MY/PH/HK/ID), delivering **10–30% incremental revenue and operational cost savings**.
- **Data Architecture:** Architected **“Data Warehouse 2.0” on Databricks**; migrated to a governed native catalog with CDC streaming via AWS DMS, **slashing data platform costs by 80%**.
- **Company-Wide AI Transformation:** Designed a git-based knowledge base and a governed **MCP layer** connecting AI agents to live company data (Databricks, Jira, GitHub, Drive, Gmail, Mixpanel).
- **AI Governance:** Implemented read-only guardrails, PII ringfencing, and infrastructure guardrails while rolling out **Claude and GPT/Codex org-wide**.
- **AI Agent Automation:** Built a **self-healing LLM pipeline** that auto-classifies Databricks job failures, self-repairs/opens PRs, and manages Jira tickets, eliminating manual triage.
- **Clinical GenAI:** Deployed patient-facing **“AI Triage”** and doctor-facing **“Clinical Support”** bots using LangChain/LangGraph on AWS AgentCore, validated via in-house eval datasets.
- **Self-Serve Analytics:** Integrated **Databricks Genie** with an MCP-connected agent for natural-language querying, reducing ad-hoc data requests from non-technical teams.
- **Predictive Modeling:** Deployed **quantile regression** for doctor demand forecasting, an **AI Fraud/Waste/Abuse detection engine** across 14 risk categories, and **B2B corporate segmentation** models.

Data Scientist | Sea Limited – Garena, Singapore Nov 2021 – Sep 2022

- Built recommendation engines for BOOYAH! (gaming social app): personalised video and community recommendations using deep neural networks (TensorFlow) on GCP over 15M+ daily samples.
- Lifted average video view duration per user **80%** and club click-through **50%**, improving engagement, growth, and retention.

Research Engineer | A*STAR – SIMTech, Singapore Aug 2014 – Aug 2016

- Led data-mining research for supply chain and manufacturing; built demand-forecasting and text-mining solutions and published **6 IEEE Big Data papers**.

EDUCATION

M.Eng. (Research), Computer Science & Design | Singapore University of Technology and Design (SUTD) 2016 – 2021
CAP 5.0 / 5.0. Thesis: Analyzing Deep Learning Models for Traffic Prediction.

B.Eng., Industrial & Systems Engineering | National University of Singapore (NUS) 2010 – 2014
CAP 4.32 / 5.0 (2nd Upper Honours). Dean’s List (Top 5%).

PUBLICATIONS

9 peer-reviewed publications in IEEE Big Data, ICPR, and ICDM, including work on integrated-gradients attribution for deep neural networks. Full list at garygsw.github.io.